

Topics -

A quick revision (session API , Entity states , dirty checking)

Continue with -

Image handling in DB

Advanced Hibernate

Inheritance

E-R (one to many n many to one : bi dir association n other association mapping)

1. Answer this

1.1 Name n explain Entity states

Transient : NO NO (exists only in java heap)

Persistent : transient -> persistent

YES NO (gains DB identity upon commit)

doesn't exist(heap) -> persistent

YES YES

Detached : NO YES

Removed : YES YES (upon commit - DML - delete , L1 cache destroyed)

NO NO -> transient

If not returned from the DAO layer method to the caller -> marked for GC

1.2 Triggers for these transition

Does Not Exist in the heap --> Transient : new

Transient --> Persistent : save | persist | saveOrUpdate | merge

Persistent --> Detached : close | evict | clear

Persistent --> Removed : delete

Detached --> Persistent : update | saveOrUpdate | merge

2. Discuss save vs persist vs saveOrUpdate vs merge

Observe n conclude

(refer to a diagram)

3. What will hibernate do upon commit , (while using getCurrentSession)

1. session.flush()

2. Hibernate performs auto dirty checking -- it will try to sync up state L1 cache with that of the DB, in case of the change , it will fire DML

new entity --insert

state updated -- update

removed entity --delete

3. session.close()

3.1 pooled out DB cn rets to DBCP

3.2 L1 cache will be destroyed.

4. What will hibernate do upon rollback ?

1. session.close()

1.1 pooled out DB cn rets to DBCP

1.2 L1 cache will be destroyed.

5. Persistent vs Detached

1. Persistent entities are stored in L1 cache.

Detached entities are not !

2. If you modify the state of the persistent entity (setters) , hibernate tracks these changes to DB (update)
3. If you modify the state of the detached entity (setters) , hibernate DOES NOT track these changes to DB (update)
4. Save n restore images to / from DB
(BLOB handling)

For easier file handling , use FileUtils class , from Apache's commons-io.jar

We have already added this dependency in pom.xml .

```
<!-- https://mvnrepository.com/artifact/commons-io/commons-io -->
```

```
<dependency>  
  <groupId>commons-io</groupId>  
  <artifactId>commons-io</artifactId>  
  <version>2.11.0</version>  
</dependency>
```

Methods of FileUtils class -

```
1. public static byte[] readFileToByteArray(File file)  
    throws IOException
```

Reads the contents of a file into a byte array n cloess the file.

```
2. public static void writeByteArrayToFile(File file,byte[] data)  
    throws IOException
```

Writes a byte array to a file creating the file if it does not exist.

File : java.io.File class , represents an abstract path to a file or the folder.

Ctor

File(String path)